

There Will Be Exports and Licensing: The Effects of Patent Rights on Firm Sales

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The statistical analysis of firm level data on U.S. multinational companies and their foreign affiliates was conducted at the Bureau of Economic Analysis, United States (U.S.) Department of Commerce under arrangements that maintain legal confidentiality requirements. Views expressed in this paper are those of the authors and do not necessarily reflect official positions of the U.S. Department of Commerce.

Outline

- ◉ Motivation
- ◉ Measurement
- ◉ Framework
- ◉ Evidence

I. Motivation

◉ Intellectual Property Rights

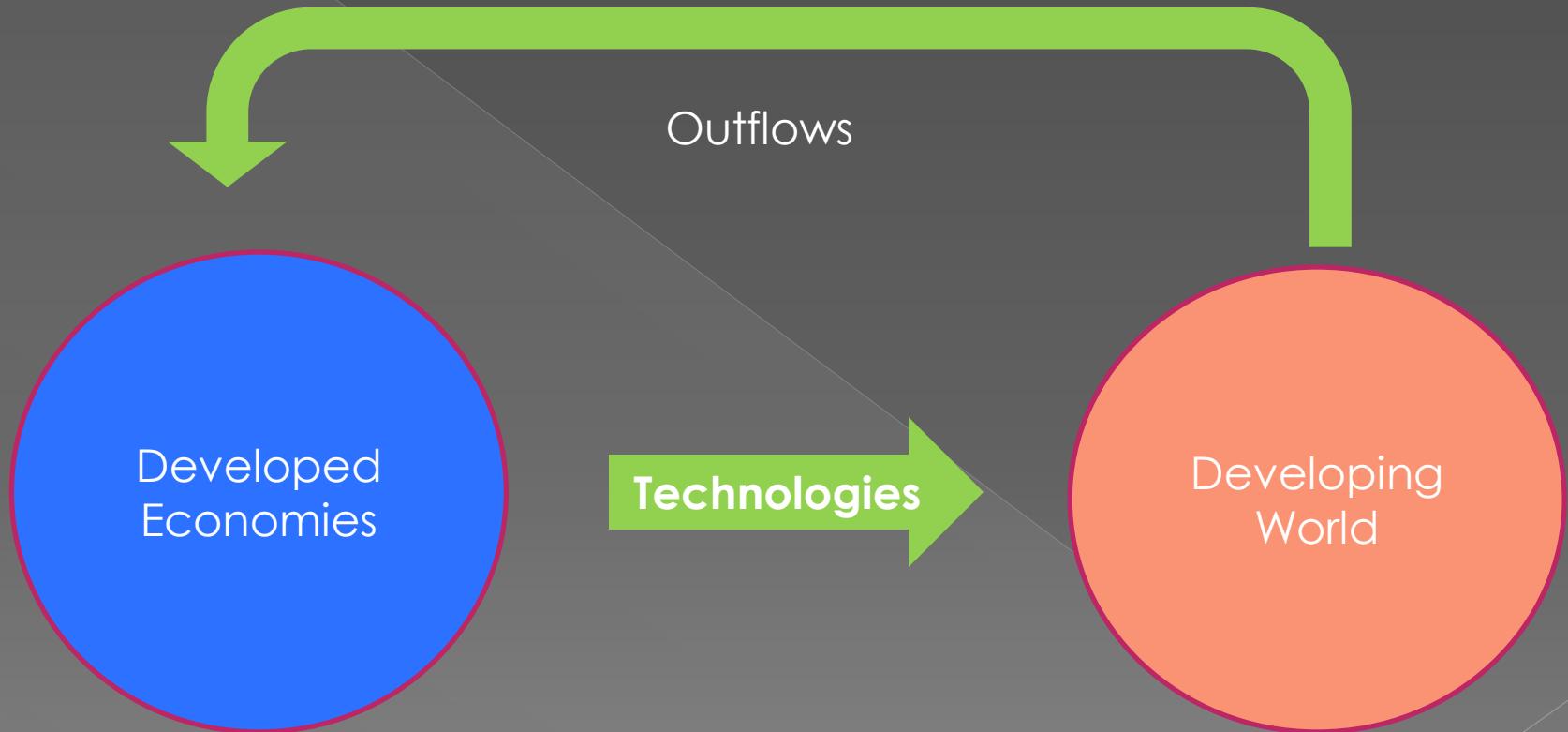
> National Context

- Innovation
- Competition Concerns

> International Context

- Non-tariff barriers
- Technology Transfer
- Economic Development
- Access

Previous Studies



Previous Studies

◉ Imports and IPRs

- > Maskus and Penubarti (1995)
- > Primo Braga and Fink (1999)
- > Smith (1999)
- > Rafiquzzaman (2002)
- > Co (2004)
- > Ivus (2010)

◉ Inward FDI/Licensing and IPRs

- > Smith (2001)
- > Yang and Maskus (2001)
- > Nunnenkamp and Spatz (2004)
- > Park and Lippoldt (2005)
- > Nicholson (2007)
- > Branstetter, Fisman, Foley, and Saggi (2011)

Previous Studies

● IPR and Innovation

- > Horwitz and Lai (1996)
- > Kanwar and Evenson (2003)
- > Chen and Puttitanun (2005)
- > Schneider (2005)
- > ... More

● Innovation, R&D, Productivity, and Exporting

- > Melitz (2003)
- > Bernard and Jensen (2004)
- > Helpman, Melitz, and Yeaple (2004)
- > Lachenmaier and Woessmann (2006)
- > MacGarvie (2006)
- > Lileeva and Trefler (2010)
- > Aw, Roberts, and Xu (2011)
- > ... More

Stylized Trends

Affiliates in Developed Countries	1982	1989	1994	1999	2004
Domestic Sales Only	27.7	27.2	32.1	38.2	32.4
Exports & Domestic Sales	54.3	55.4	49.5	43.8	50.1
Outward Licensing	17.9	17.4	15.1	18.0	17.5

Affiliates in Developing Countries	1982	1989	1994	1999	2004
Domestic Sales Only	40.5	41.4	40.0	48.2	31.0
Exports & Domestic Sales	57.2	54.8	56.0	46.4	63.8
Outward Licensing	2.3	3.8	4.0	5.4	5.2

% of firms with respective activity

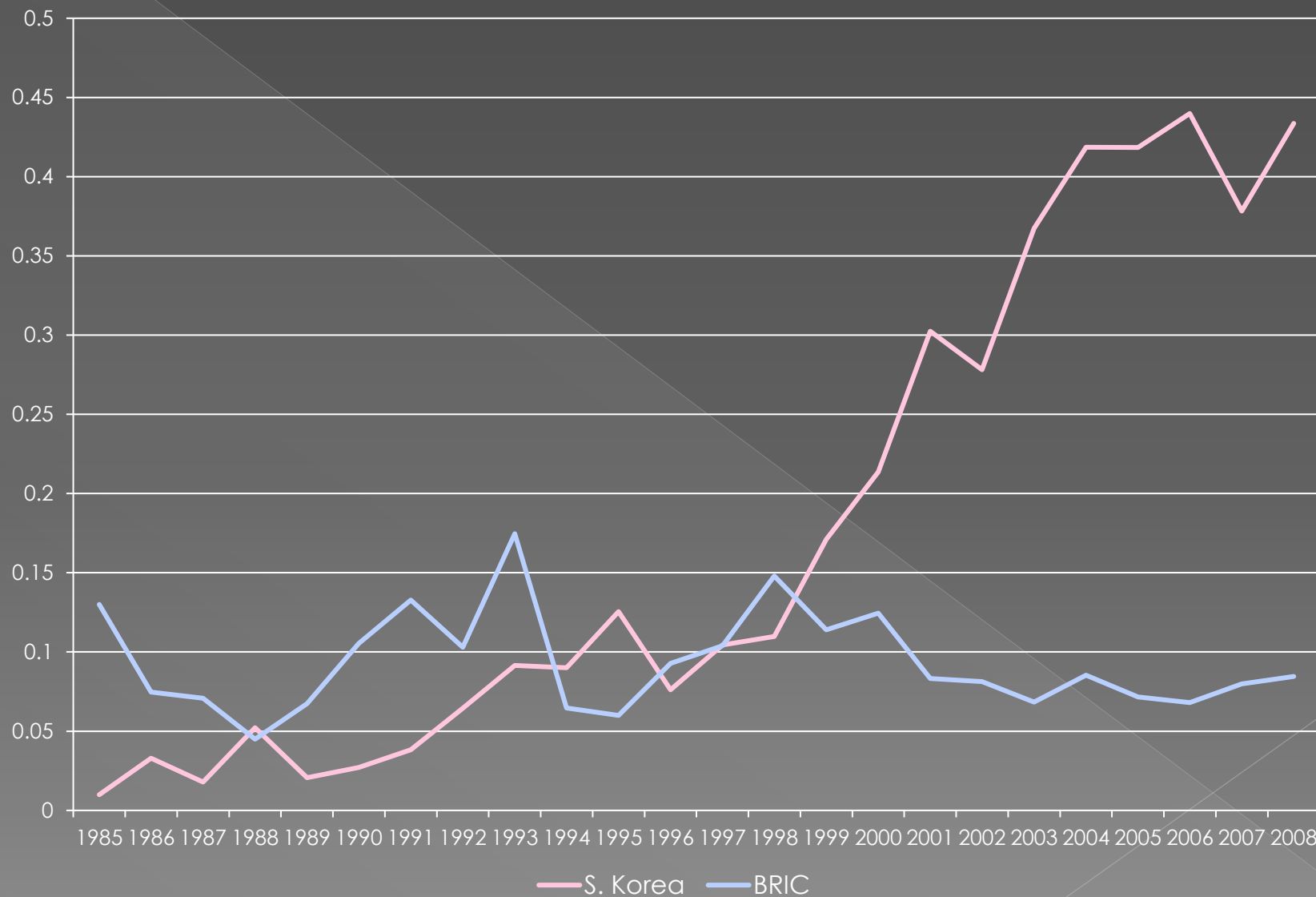
Stylized Trends

South Korea	1982	1989	1994	1999	2004
Domestic Sales Only	57.1	50.0	51.9	48.1	38.9
Exports & Domestic Sales	42.9	42.9	38.9	40.7	50.6
Outward Licensing	0.0	7.1	9.2	11.1	10.5

China	1982	1989	1994	1999	2004
Domestic Sales Only	n/a	88.9	41.9	50.2	25.6
Exports & Domestic Sales	n/a	11.1	55.8	48.8	69.7
Outward Licensing	n/a	0.0	2.3	1.0	4.7

% of firms with respective activity

RATIO of Licensing Receipts to Licensing Payments (Balance of Payments)



Patents Granted at U.S.P.T.O

	Rank 2010	Rank 2004
Japan	1	1
Germany	2	2
S. Korea	3	4
Taiwan	4	3
Canada	5	7
France	6	6
U.K.	7	5
China	8	19
Israel	9	12
Italy	10	8

II. Measurement

◉ Patent Rights Index

- > U.S. and International Chamber of Commerce Guidelines (1988)
- > Statutes and Case Laws
- > Basis for Patent Disputes & Litigation
- > Variability

II. Measurement

◉ Patent Rights Index (0 - 5)

- Duration (0 - 1)
- Coverage (0 - 1)
- Restrictions, if any (0 - 1)
- Enforcement Mechanisms (0 - 1)
- Membership in International Treaties (0 - 1)

Sample:

Year	U.S.A.	S. Korea
1985	4.68	2.65
1990	4.68	3.69
1995	4.88	3.89
2000	4.88	4.13
2005	4.88	4.33

Alternative Measures

Correlation with Patent Rights Index (Coefficient of Variation = 0.47)

Other Measures:	1995	2000	2005	Coefficient of Variation
Economist Intelligence Unit	0.71	0.73	0.72	0.39
World Economic Forum	n/a	0.74	0.67	0.33
Fraser Institute	0.69	0.66	0.68	0.34
Copyright Index	0.60	0.65	0.65	0.30
Trademark Index	0.60	0.50	0.59	0.33

III. Framework

- Firm has four choices:
 - > Sell domestic only
 - > Sell domestic and Export
 - > Sell domestic and License
 - > Sell domestic, Export, and License
- Decisions $\theta_{it} = \{\text{Domestic Sales } \theta_{it}^D, \text{ Export Sales } \theta_{it}^X, \text{ Licensing Sales } \theta_{it}^L\}$

Value Function

$$V_{it}(s_{it}) = \max_{\{\theta_{it}\}} (\pi_{it}^D + \max \{V_{it}^D(s_{it}), \hat{\pi}_{it}^X + V_{it}^X(s_{it}), \hat{\pi}_{it}^L + V_{it}^L(s_{it}), \hat{\pi}_{it}^X + \hat{\pi}_{it}^L + V_{it}^{XL}(s_{it})\})$$

where $s_{it} = (\omega_{it}, x_{it-1}, l_{it-1})$ is the state vector

$$\hat{\pi}_{it}^X = \pi_{it}^X - (x_{it-1} \gamma^F + (1 - x_{it-1}) \gamma^S)$$

$$\hat{\pi}_{it}^L = \pi_{it}^L - (l_{it-1} \phi^F + (1 - l_{it-1}) \phi^S)$$

and V_{it}^m , $m = \{D, X, \text{ and } L\}$ defined next $\rightarrow$

$$V_{it}^D(s_{it}) = \delta E_t V_{it+1}(s_{it+1} \mid \mathbf{x}_{it} = 0, l_{it} = 0)$$

$$V_{it}^X(s_{it}) = \delta E_t V_{it+1}(s_{it+1} \mid \mathbf{x}_{it} = 1, l_{it} = 0)$$

$$V_{it}^L(s_{it}) = \delta E_t V_{it+1}(s_{it+1} \mid \mathbf{x}_{it} = 0, l_{it} = 1)$$

$$V_{it}^{XL}(s_{it}) = \delta E_t V_{it+1}(s_{it+1} \mid \mathbf{x}_{it} = 1, l_{it} = 1)$$

$$\pi_{it}^m = \pi(\omega_{it}, \theta_{it}, \Phi_{it}, .), \quad m = \{D, X, L\}$$

$$\Phi_{it} = \Phi(k_{it}, r_{it}, IPR_t, .)$$

- Productivity

$$\omega_{it} = g(\omega_{it-1}, r_{it-1}, x_{it-1}, l_{it-1}) + \xi_{it}$$

- Solution to Value Maximization Problem (1):

$$X_{it} = X(s_{it}, \Phi_{it}, .)$$

$$L_{it} = L(s_{it}, \Phi_{it}, .)$$

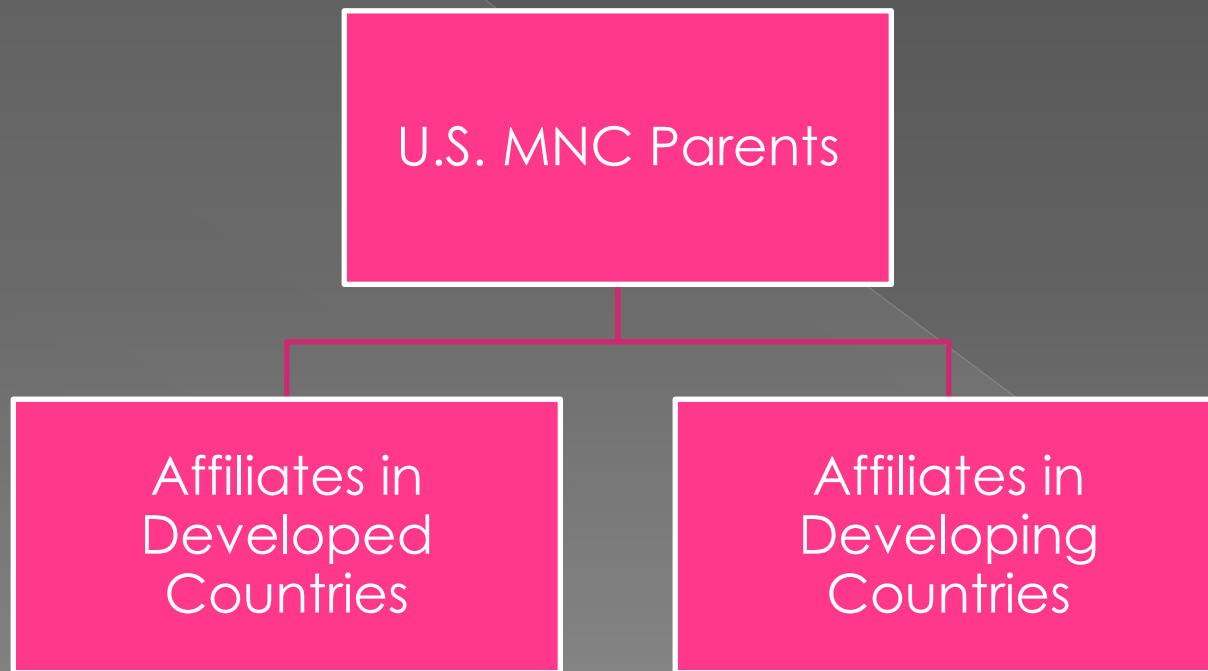
where $s_{it} = (\hat{\omega}_{it}, x_{it-1}, l_{it-1})$

Digression: 1st stage estimates of productivity

- Follow methodology of Aw et al (2011), Levinsohn and Petrin (2003), Doraszelski and Jaumandreu (2008)
- Basic Steps:
 - > Locate Productivity in $R_{it} = R(k_{it}, \omega_{it}) + u_{it} = R^1(k_{it}) + R^2(\omega_{it}) + u_{it}$
 - > Assume $\omega_{it} = \omega(z_{it} \mid k_{it})$, where Z are observables
 - > Substituting, $R_{it} = h(k_{it}, z_{it}) + u_{it}$
 - > Run OLS and get $\hat{h}_{it}$
 - > Match $\hat{h}_{it}$ to $R(k_{it}, \omega_{it})$ and “back out” $\omega_{it} = \varpi(k_{it}, \hat{h}_{it})$
 - > Estimate parameters in backed out equation, ...
 - > Yields $\hat{\omega}_{it}$

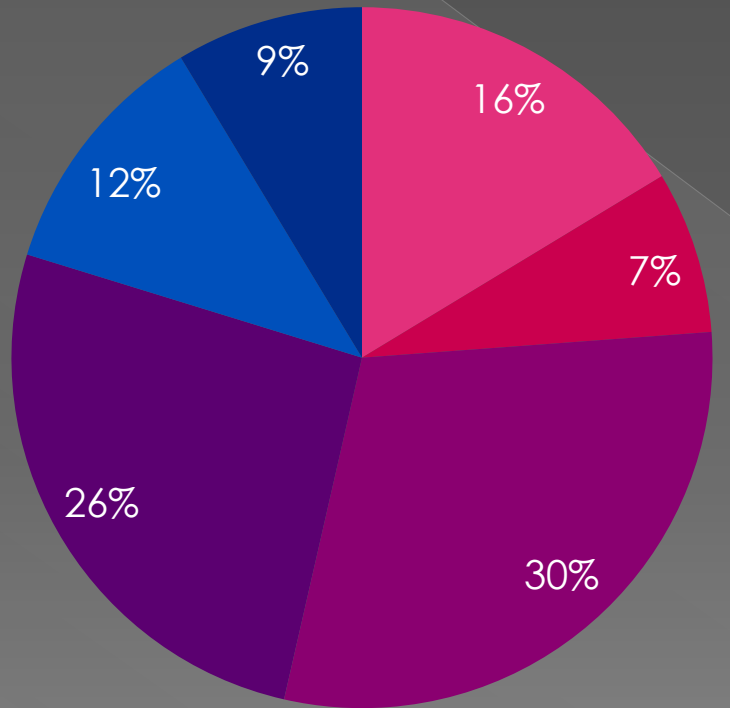
IV. Evidence

◉ Micro-Data



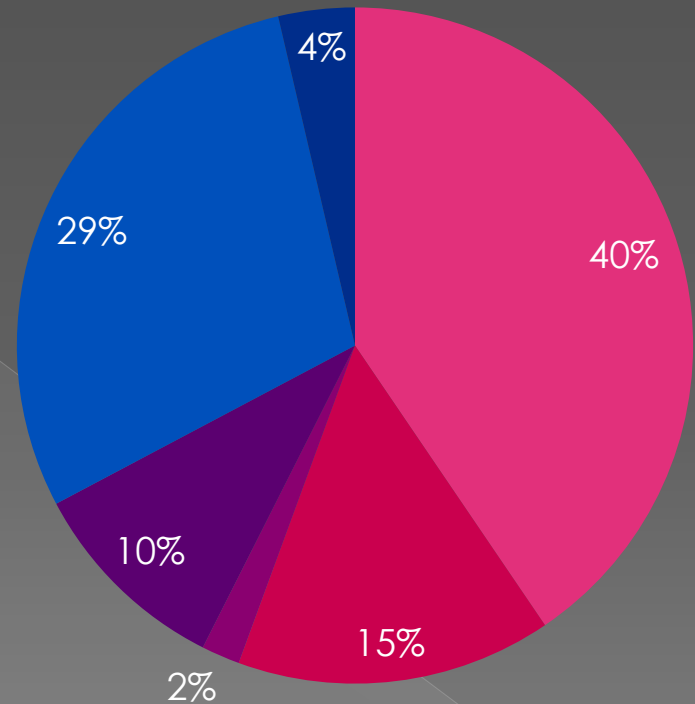
Composition of Firms

Sector



■ Chemicals
■ Machinery
■ Services
■ Communications
■ Retail, Wholesale
■ Agriculture and other

Region



■ Europe
■ Africa
■ N. America
■ Asia
■ Latin America
■ Other

Switching between Modes

● Aggregate Data



● Micro Data



Overview:

- ◉ Key Objective: IPR and Outward Orientation
- ◉ Three-Prong Approach:
 - > Bivariate System and IPR
 - > Ordered Response and IPR
$$\phi^F + \phi^S > \gamma^F + \gamma^S$$
 - > Affiliate vs. Unaffiliated Licensing and IPR

Preliminary: Endogeneity Test

- Between IPR and Firm Exports and Licensing
- Instruments:
 - > Governance Indicators, Legal Origins, Political Rights
- F-test of Excluded Regressors:
 - > Regression of IPR on Included & Excluded Regressors (p-value = 0.000)
- Overidentification Test:
 - > $\chi^2(4)$ p-values = 0.62 for Export, and = 0.14 for Licensing
- Durbin-Wu-Hausman Test:
 - > $\chi^2(1)$ p-values = 0.22 for Export, and = 0.38 for Licensing
- Cannot Reject NULL of Exogeneity of IPR

[illegible]

Sub-samples

[illegible]

II. Ordered Probit Analysis

- Latent variable $y^* = (\text{vector of explanatory factors})'\gamma + \varepsilon$
- $y = 1$ (No Exporting, No Licensing) if $y^* \leq \text{Threshold 1}$
- $y = 2$ (Exporting, No Licensing) if $\text{Threshold 1} \leq y^* \leq \text{Threshold 2}$
- $y = 3$ (Licensing) if $y^* \geq \text{Threshold 2}$

II. Ordered Probit Analysis

RHS in logs \	Developed Countries	Developing Countries
IPR	-0.246	0.291***
R&D	0.079***	0.054***
K	0.066***	0.124***
Productivity	0.647***	0.843***
GDP	0.066**	0.010
Threshold 1	4.612***	3.864***
Threshold 2	6.639***	5.965***
No. of Observ.	9743	6787
Constant, Dummies for Time, Region, Country, Industry, and Trade Agreements		

Marginal Effects from Ordered Probit Estimates

A. U.S. Parent Firms and Foreign Affiliates in Developed Countries

	Outcome #1:	Outcome #2:	Outcome #3:
	No Exports or	Exports	Licensing
Variable	Licensing	(No Licensing)	(with or without Exports)
In Patent Rights	-0.076	0.018	0.058
In R&D	-0.025***	0.006***	0.019***
In Productivity	-0.200***	0.047***	0.154***
Predicted Probabilities	23.9%	60.7%	15.4%
Actual Frequencies	31.5%	50.6%	17.9%

B. Foreign Affiliates in Developing Countries

	Outcome #1:	Outcome #2:	Outcome #3:
	No Exports or	Exports	Licensing
Variable	Licensing	(No Licensing)	(with or without Exports)
In Patent Rights	-0.105***	0.076***	0.029***
In R&D	-0.019***	0.014***	0.005***
In Productivity	-0.306***	0.222***	0.084***
Predicted Probabilities	33.1%	62.0%	4.9%
Actual Frequencies	39.1%	56.8%	4.1%

Notes:

***, **, and * indicate significance levels of 1%, 5%, and 10% respectively.

Calculations are based on columns (2) - (3) of Table 5.

	All Countries		Developed		Developing	
RHS in logs \	1. Ln Affiliated	2. Ln Unaffiliated	3. Ln Affiliated	4. Ln Unaffiliated	5. Ln Affiliated	6. Ln Unaffiliated
IPR	2.38**	3.58***	-5.45	-3.15	1.43	4.54***
R&D	0.48***	0.51***	0.47***	0.54***	0.31***	0.29***
K	0.48***	0.40***	0.46***	0.34***	0.60***	0.71***
Productivity	3.14***	7.12***	3.09***	7.65***	2.65*	6.20***
GDP	0.41***	0.38**	0.70***	0.76***	-0.43	0.13
Past Affiliated Licensing	0.31	11.03***	0.09	9.32***	1.01	16.1***
Past Unaffiliated Licensing	9.61***	3.87***	9.54***	3.38***	11.20***	4.61***
Past Affil x Past Unaffil	-0.107	-5.10***	-1.43*	-3.68***	0.07	-6.20***
ρ	0.504***		0.489***		0.412***	
Other: Constant and Dummies for Time, Country, Region, Industry, and Trade Agreements						

Conclusion: Highlights

- Built on literature on Innovation, Productivity, and Trade
- IPR part of outward oriented strategy
- IPR → Innovation, Contracting → Facilitates Trade
- Firm-level data: can detect behavior that more aggregative data cannot
- Critical Level of IPR:
 - Local Sales → Exporting → Licensing to Affiliates → Licensing to Arms-length parties